

CIS2987 CIS Capstone

Project Proposal 1

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Smart Pet Water Fountain Prototype and Infrastructure for Pet-Technology Startup (Name TBD)

A new pet-technology startup (company name TBD) is developing an automated smart water fountain designed to provide continuous, filtered hydration for household pets. The company's goal is to create a reliable, sensor-driven device that can operate autonomously while also offering optional remote control through a web or mobile interface. As the founder and lead developer, I will design and implement the first functional prototype of the device along with the supporting digital infrastructure.

The company currently has no technical systems in place. There is no microcontroller firmware, no automation logic, no website, and no secure online presence. To move toward product launch, the startup requires a working hardware prototype, a secure informational website, and preliminary IoT control capabilities.

Scope of Work

Hardware & Firmware Development

- Program a microcontroller (Arduino or Raspberry Pi Pico) to control pump cycles, water-level detection, and automated operation.
- Integrate sensors as needed for safe and efficient operation.
- Test and refine the automation logic to ensure reliability.

Web Infrastructure

- Register and configure a domain for the startup (name TBD).
- Deploy a secure website using HTTPS, basic hardening, and clean navigation.
- Create pages describing the product, features, documentation, and company information.

Optional IoT Control Interface

- Develop a simple web-based dashboard or mobile-friendly interface to toggle fountain modes or view device status.
- Implement secure communication between the device and the control interface.

Documentation & Project Management

- Produce hardware diagrams, wiring schematics, and code documentation.
- Maintain weekly project management logs.
- Prepare a final report summarizing the system architecture, testing results, and future development recommendations.

Task Breakdown and Estimated Hours

Hardware & Firmware (18 hours)

- Microcontroller setup and environment configuration (2 hours)
- Pump control programming (3 hours)
- Sensor integration and calibration (4 hours)
- Automation logic (4 hours)
- Hardware testing and refinement (5 hours)

Web Infrastructure (12 hours)

- Domain registration and DNS configuration (1 hour)
- Web server setup and security hardening (3 hours)
- Website design and content creation (5 hours)
- HTTPS configuration and testing (3 hours)

IoT Control Interface (Optional Component) (8 hours)

- Basic dashboard or control page (3 hours)
- Communication logic (3 hours)
- Testing and refinement (2 hours)

Documentation & Reporting (8 hours)

- Hardware diagrams and wiring documentation (2 hours)
- Code documentation (2 hours)
- Website documentation (1 hour)
- Final report preparation (3 hours)

Project Management (6 hours)

- Weekly planning and milestone tracking (1 hour per week × 6 weeks)

Total Estimated Hours: 52 hours

Timeline for Completion (Weeks 9–16)

Week 9

- Set up microcontroller environment
- Begin pump control programming
- Start project management tracking

Week 10

- Integrate and calibrate sensors

- Continue firmware development
- Begin hardware testing

Week 11

- Complete automation logic
- Conduct full hardware test cycle
- Begin domain registration and web server setup

Week 12

- Implement website structure and content
- Configure HTTPS and basic security
- First progress report

Week 13

- Develop IoT control dashboard (optional)
- Implement device-to-interface communication
- Continue documentation

Week 14

- Test IoT features
- Refine website and firmware
- Update documentation and project logs

Week 15

- Finalize all hardware and software components
- Conduct full system testing
- Begin final report

Week 16

- Complete final report
- Finalize documentation package
- Submit capstone deliverables